Materials Science 1 – Diffusion in Solids
Part 2

Learning objectives and learning outcomes of this lecture

1. Introduce the concept of non-steady state diffusion in solids;

2. Identify the factors that affect diffusion;

3. Perform calculations for non-steady state diffusion in solids using Fick’s second law;

4. Distinguish between the effects of temperature and diffusing species on the mechanism of diffusion in solids.

Poll – Answers in WeChat

Q1) As the temperature rises, what occurs to the rate of vacancy sites in metals?

A) Increases
B) Decreases
C) Remains the same

Poll – Answer

Q1) As the temperature rises, what occurs to the rate of vacancy sites in metals?

A) Increases

Since more energy is available at higher temperature, it is easier to break bonds and form additional vacancies.

Fick's second law and non-steady state conditions

In steady state conditions, the concentration profile and the diffusion flux are constant over time.

In a situation where the diffusion flux along a direction x is not constant (i.e. mass of diffusing species entering the volume region is higher than the mass exiting the volume region), there will be accumulation of mass within the volume. This produces a time-varying concentration, expressed by the Fick's second law. Assuming that the diffusion coefficient is constant along the x direction, the Fick's second law can be written as:

Fick's second law

$$\boxed{\frac{dC}{dt} = D_{AB}\frac{d^2C}{dx^2}}$$

This is a partial differential equation, also known as diffusion equation and has for solution a function of the type C=C(x, t).

Note that the Fick's second law can be easily derived by continuity equation (mass conservation condition)

Assuming again a one-dimensional system where now **concentration changes over time** as well as due to diffusion but the fluid is still at rest

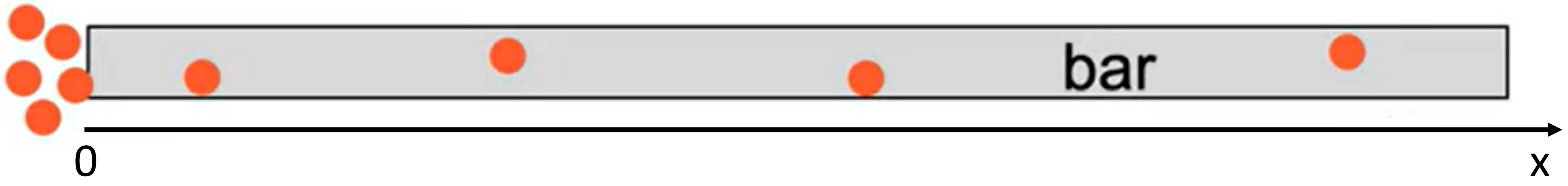
$$\frac{\partial c}{\partial t} + \frac{\partial J}{\partial x} = 0$$

one yields with Fick's first law

$$\frac{\partial c}{\partial t} - \frac{\partial}{\partial x}\left(D\frac{\partial c}{\partial x}\right) = 0$$

Example of solution of Fick's second law

Consider the case of Cu particles diffusing into an aluminium long (semi-infinite) bar:



The Fick's second law allows to determine the concentration of Cu as a function of time and position along the bar, given the initial and boundary conditions.

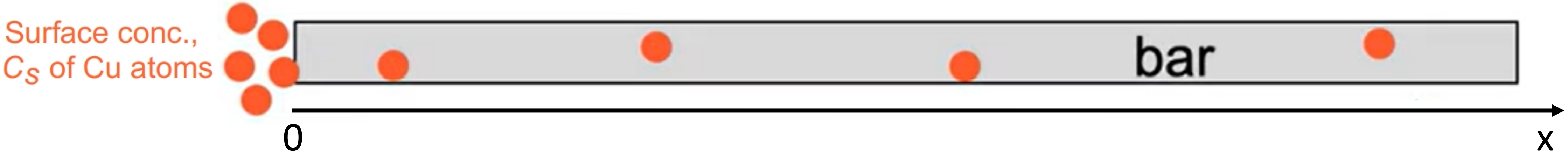
For the initial and boundary conditions, let us pose:

at $t = 0$, $C = C_o$ for $0 \leq x \leq \infty$ *(initial conditions)*

at $t > 0$, $C = C_S$ for $x = 0$ *(constant surface conc.)*

$C = C_o$ for $x = \infty$ *(conc. at infinity)*

Example of solution of Fick's second law



With the initial and boundary conditions we have imposed, the solution indicates that the concentration at any position x along the bar increases with time. Meanwhile, the concentration will decrease with increasing x:

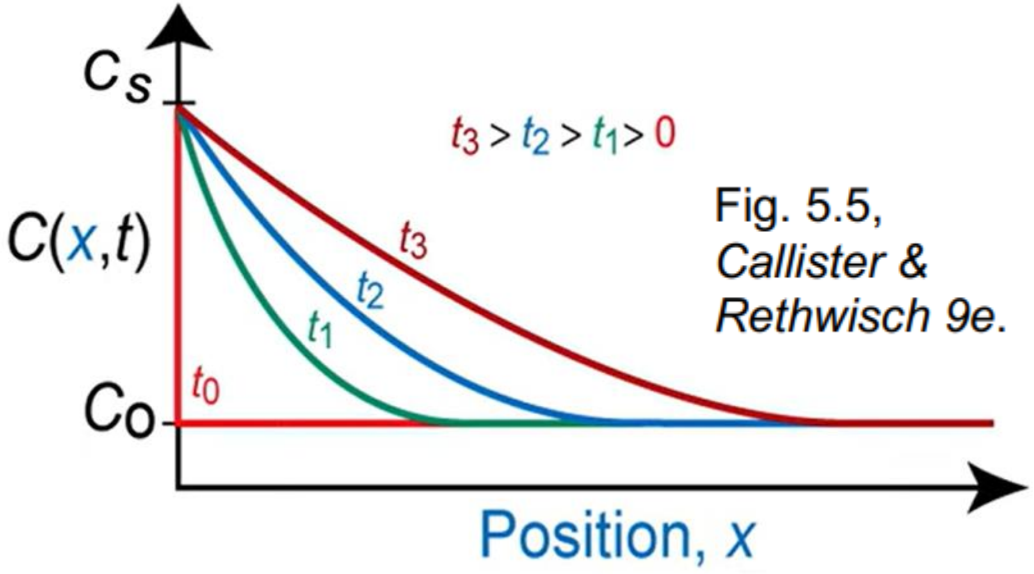


Fig. 5.5, *Callister & Rethwisch 9e.*

Mathematically, the function C(x,t) is obtained from the following relationship:

$$\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \mathrm{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Example of solution of Fick's second law – the Erf function

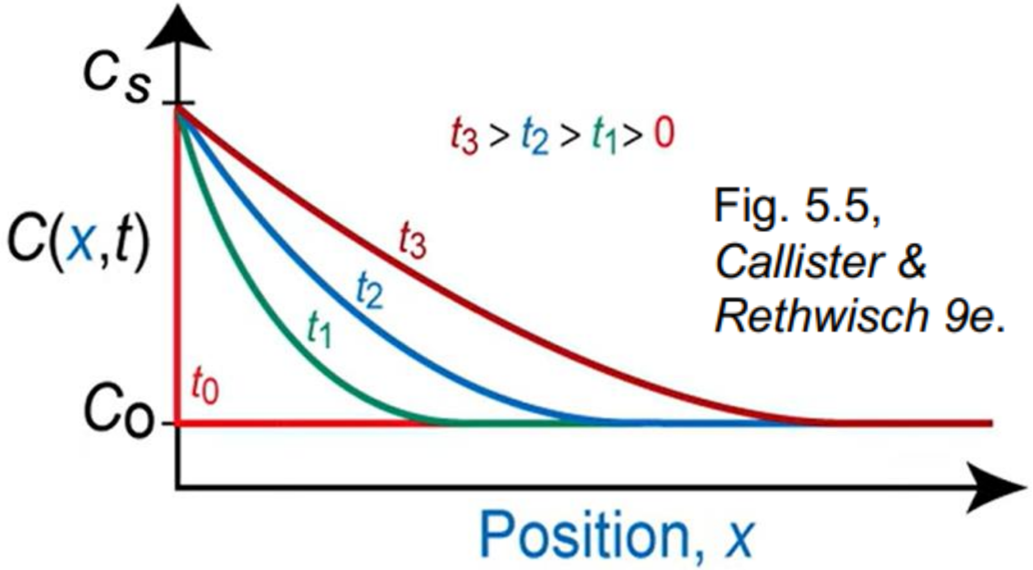


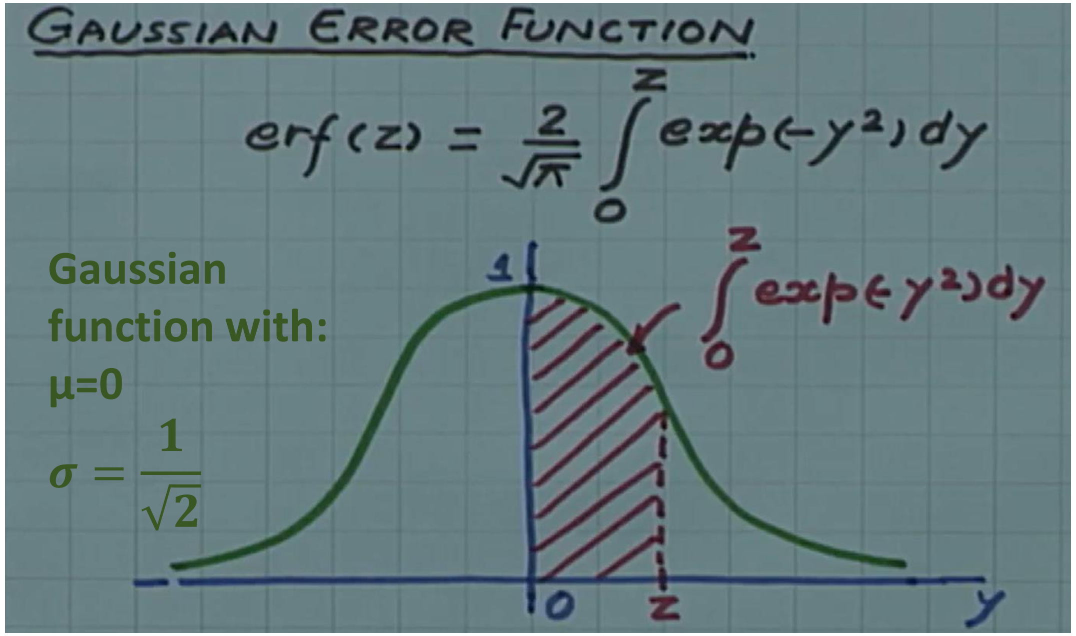
Fig. 5.5, *Callister & Rethwisch 9e.*

$$\boxed{\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)}$$

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mathematically, the function erf(z) is called the **error function** (or Gauss error function), and is defined as:

$$\boxed{\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-y^2}\,dy}$$



For more details on the error function, a link to a video will be uploaded on QM+.

Error function tabled values

z	$erf(z)$	z	$erf(z)$	z	$erf(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

- Erf(z) is the error function;
- The value can be obtained from tables;
- If z/erf(z) is in between values of two entries in the table, we use interpolations.

Why is it called "Error function"?

- The name comes from the general theory of errors in measurements, according to which the entity of error of a specific observation can be regarded a variable with a Gaussian distribution.

- When the results of a series of measurements are described by a normal distribution with standard deviation σ, the error function:

$$erf\left(\frac{a}{\sigma\sqrt{2}}\right)$$

expresses the probability that the error of a single measurement is between –a and a.

Applications of Fick's second law

Example 1: Non-steady State Diffusion of N in Fe

C_s

Nitrogen (N) from a gaseous phase is to be diffused into pure iron (Fe) at 675 °C. If the surface concentration is maintained at 0.2 wt% N, determine:

$\rightarrow C$

- What will be the concentration of N 2mm from the surface after 25h?

The diffusion coefficient for N in Fe at 675 °C is 1.9 x 10^{-11} m^2/s D

$$\frac{dC}{dt} = D_{AB}\frac{d^2C}{dx^2}$$

$$\frac{C_1 - 0.2wt\%}{25\times3600s} = 1.9\times10^{-11}$$

① 先算 erf( ) 里

② 表格找数

$$\frac{C(x,t) - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

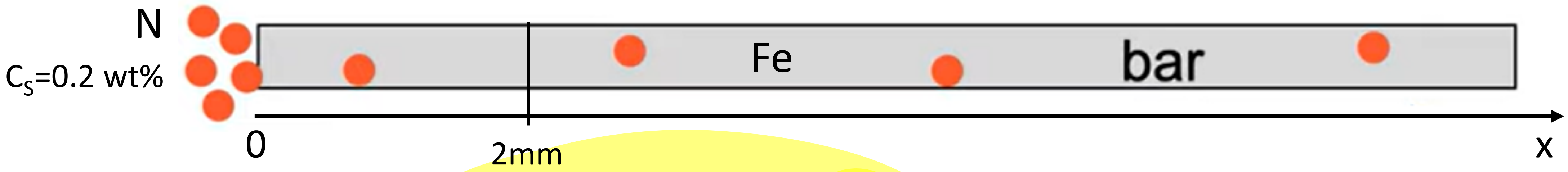
Applications of Fick's second law

<u>Example 1 solution:</u> Non-steady State Diffusion of N in Fe

Nitrogen (N) from a gaseous phase is to be diffused into pure iron (Fe) at 675 °C. If the surface concentration is maintained at 0.2 wt% N, determine:

- What will be the concentration of N 2mm from the surface after 25h?

The diffusion coefficient for N in Fe at 675 °C is 2.8 x 10^{-11} m^2/s.



D at 675 °C = 2.8 x 10^{-11} m^2/s
C_0 = 0 (pure Fe) No N initially
C_s = 0.2 wt%
x = 0.002m
t = 25h = 90000s

$$\frac{C(x,t)-C_o}{C_s-C_o} = 1-\text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Estimating the value of erf:

$$1 - \text{erf}\left[\frac{2 \times 10^{-3}\ \text{m}}{(2)\sqrt{(2.8\times10^{-11}\ \text{m}^2/\text{s})(25\ \text{h})(3600\ \text{s/h})}}\right] = 1-\text{erf}\,(0.630)$$

Error function tabled values

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z	erf(z)	z	erf(z)	z	erf(z)
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
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0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

- z=0.63 is not present in the table. Therefore, we need to use interpolation:

Finding erf(z) by interpolation from the table:

z	erf (z)
0.600	0.6039
0.630	y
0.650	0.6420

$z' - z_1$ erf − erf(z_1)

$$\frac{0.630 - 0.600}{0.650 - 0.600} = \frac{y - 0.6039}{0.6420 - 0.6039}$$

$z_2 - z_1$ erf(z_2) −

$$y = \text{erf}\,(0.630) = 0.6268$$

$$\frac{C_x - 0}{0.2 - 0} = 1.0 - 0.6268$$

And solving for *Cx* gives: *Cx* = 0.075 wt% N

Concentration of N in Fe at 2mm after 25h.

Applications of Fick's second law

Example 2: Non-steady State Diffusion of C in Fe

t

Determine the carburizing (diffusion) time necessary to achieve a carbon concentration of 0.30 wt% at a position 4 mm into an iron–carbon alloy that initially contains 0.10 wt% C. The surface concentration is to be maintained at 0.90 wt% C, and the treatment is to be conducted at 1100°C. The diffusion of carbon into γ-iron data are as follows: D_0 = 2.3 x 10^{-5} m^2/s, Q_d = 148,000 J/mol.

$C_s = 0.9$ $\qquad$ $C_0 = 0.1$

$$\frac{C(x,t) - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

$C(4 \times 10^{-3}, t) = 0.3$ wt%

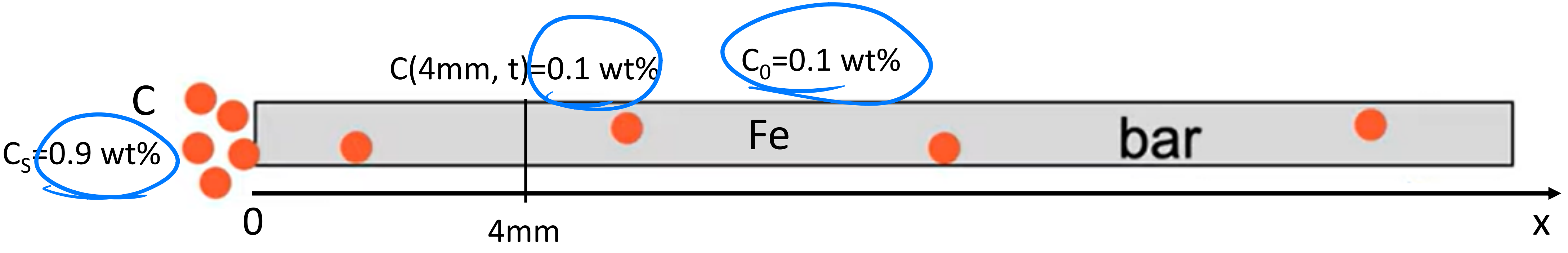
$\frac{3}{4}$

$$\frac{0.3 - 0.1}{0.9 - 0.1} = 1 - erf\left(\frac{0.004}{2\sqrt{2 \times 10^{-5} t}}\right) = 0.75$$

$$\frac{1}{4} = 1 - erf$$

Applications of Fick's second law

Example 2 solution:



Solution of Fick's second law

$$\frac{C(x,t)-C_o}{C_s-C_o}=1-\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Using the given data:

$$\frac{C_x-C_0}{C_s-C_0}=\frac{0.30-0.10}{0.90-0.10}=0.2500=1-\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)=1-0.2500=0.7500$$

Example 2 solution - Error function tabled values

z	erf(z)	z	erf(z)	z	erf(z)
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

0.80 0.7421
z 0.75
0.85 0.7707

$$\frac{z-0.800}{0.85-0.80} =$$

- Erf(z)=0.75 is not present in the table. Therefore, we need to use interpolation:

z	erf(z)
0.80	0.7421
z	0.7500
0.85	0.7707

$$\frac{z-0.800}{0.850-0.800} = \frac{0.7500-0.7421}{0.7707-0.7421}$$

$$z = 0.814 = \frac{x}{2\sqrt{Dt}}$$

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Example 2 solution – Calculating D and t

The data for the diffusion of carbon into γ-iron are as follows:

D_0 = 2.3 ×10–5 m2/s

Q_d = 148,000 J/mol

D: diffusion coffieient

?

Therefore at 1100 °C (1373 K)

$$D = (2.3 \times 10^{-5}\ \mathrm{m^2/s})\exp\left[-\frac{148{,}000\ \mathrm{J/mol}}{(8.31\ \mathrm{J/mol\text{-}K})(1373\ \mathrm{K})}\right] = 5.35 \times 10^{-11}\ \mathrm{m^2/s}$$

Substituting in the equation above $z = 0.814 = \dfrac{x}{2\sqrt{Dt}}$ $\Rightarrow$ $0.814 = \dfrac{4 \times 10^{-3}\ \mathrm{m}}{(2)\sqrt{(5.35 \times 10^{-11}\ \mathrm{m^2/s})(t)}}$

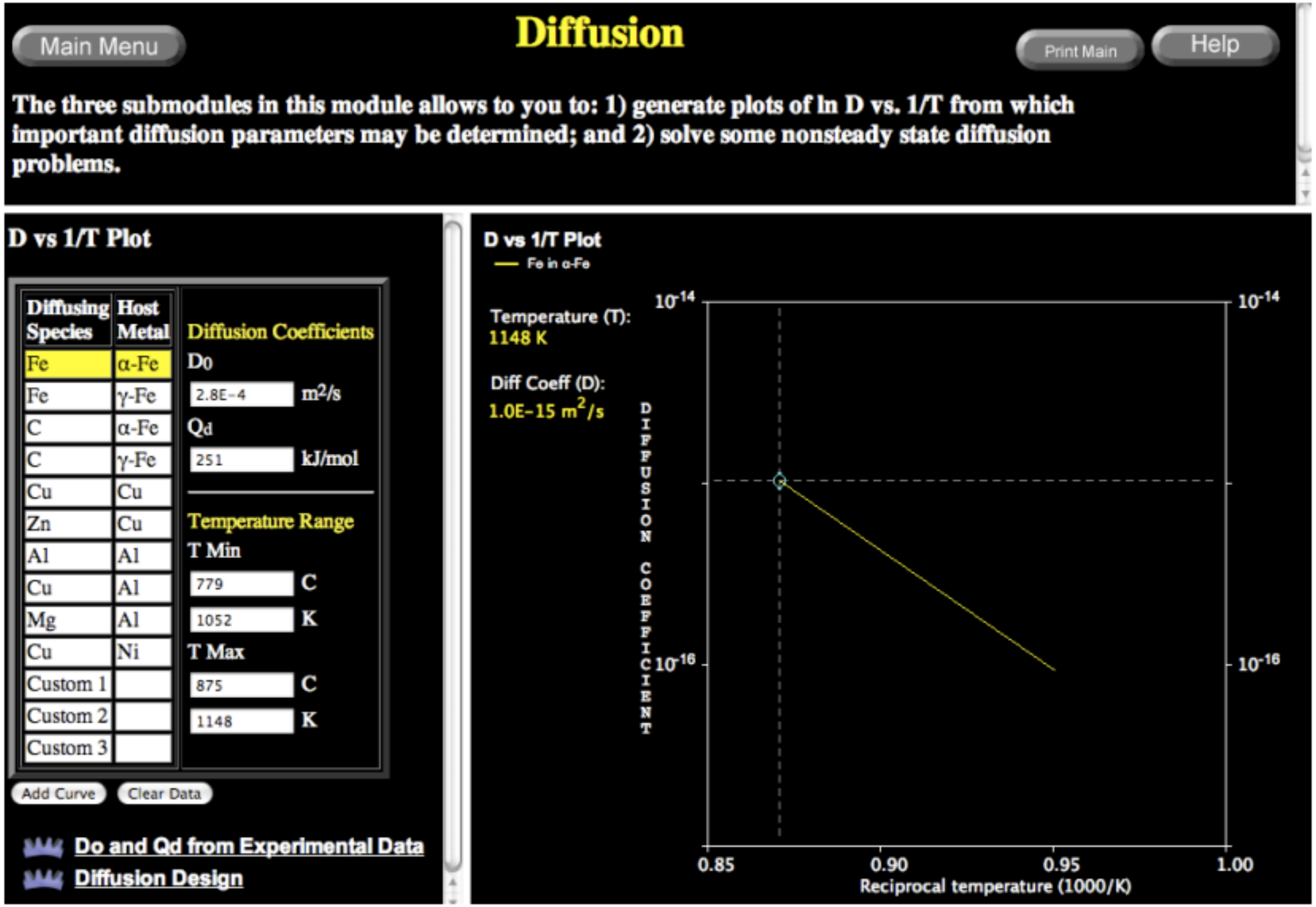
Solving for t yields
$$t = \frac{(4 \times 10^{-3}\ \mathrm{m})^2}{(2)^2(0.814)^2(5.35 \times 10^{-11}\ \mathrm{m^2/s})}$$

Therefore $t = 1.13 \times 10^5\ \mathrm{s} = 31.3\ \mathrm{h}$

It takes 31.3 h to diffuse enough C into Fe-steel to achieve a 0.30% wt at 4mm of the bar surface.

Virtual Materials Science and Engineering (VMSE) website

Screenshot of Diffusion Calculations and Data Plots



It allows estimating important parameters of diffusion processes (i.e. diffusion coefficients at different T for different materials, activation energy, etc).

Factors that affect diffusion

Diffusion and temperature

Diffusion is a thermally activate process. The diffusion coefficient increases with increasing T.

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

Arrhenius-type relationship

D = diffusion coefficient [m²/s]

D_o = pre-exponential [m²/s]

Q_d = activation energy [J/mol or eV/atom]

R = gas constant [8.314 J/mol-K]

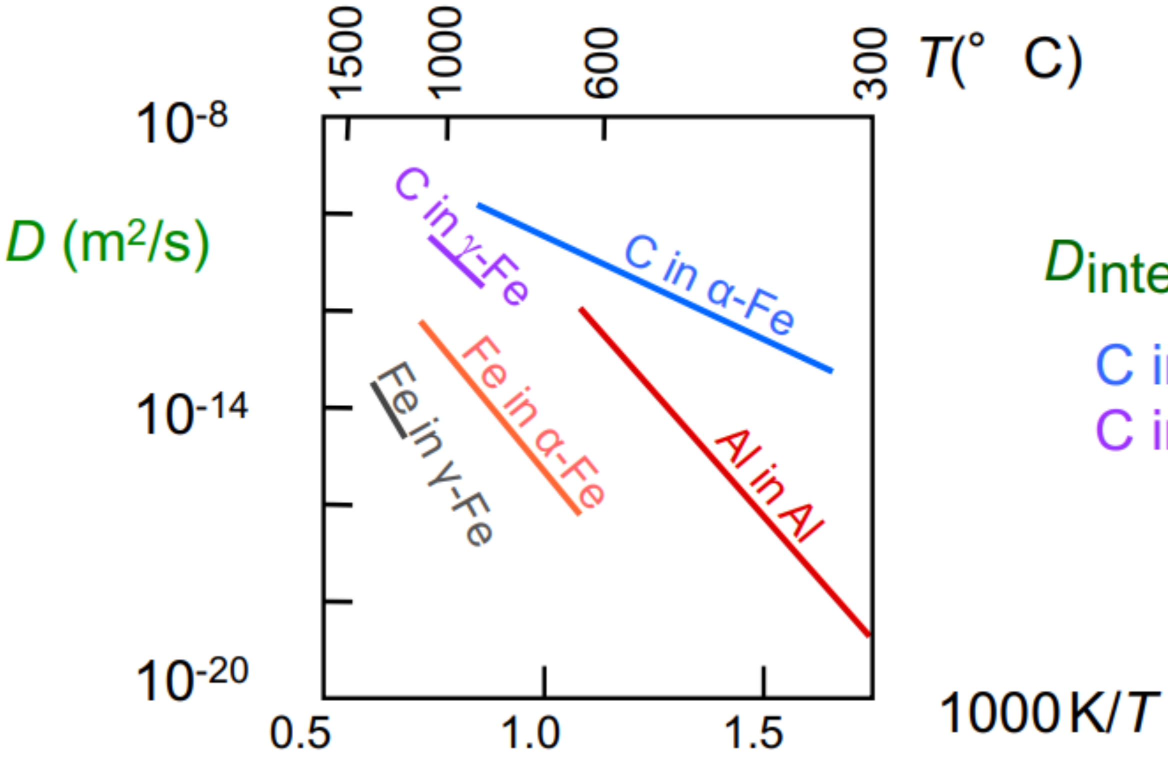
T = absolute temperature [K]

The activation energy is the energy required to produce the diffusive motion of one mole of atoms. A large activation energy results in a relatively small diffusion coefficient.

Diffusion and temperature

Taking natural logarithms, the temperature dependence of the diffusion coefficient is a linear function of $\frac{1}{T}$. The plot can be used to estimate the activation energy of the diffusion process.

$$\ln D = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T}\right)$$



$D_{interstitial} >> D_{substitutional}$

$D_{interstitial}$	$D_{substitutional}$
C in α-Fe	Al in Al
C in γ-Fe	Fe in α-Fe
	Fe in γ-Fe

Adapted from Fig. 5.7, *Callister & Rethwisch 9e.*
(Data for Fig. 5.7 taken from E.A. Brandes and G.B. Brook (Ed.) *Smithells Metals Reference Book*, 7th ed., Butterworth-Heinemann, Oxford, 1992.)

Diffusion and temperature

Example 1: Estimation of diffusion coefficient at specific temperature

At 300°C the diffusion coefficient and activation energy for Cu in Si are D(300°C) = 7.8×10^{-11} m^2/s and Q_d = 41.5 kJ/mol. What is the diffusion coefficient at 350°C?

Diffusion and temperature

Example 1 solution:

Write the temperature dependence of diffusion coefficient in logarithmic form at T_1 and T_2:

$$\ln D_1 = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T_1}\right) \qquad \ln D_2 = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T_2}\right)$$

Relate D_1 and D_2:

$$\ln D_2 - \ln D_1 = \ln\frac{D_2}{D_1} = -\frac{Q_d}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \Rightarrow D_2 = D_1 \exp\left[-\frac{Q_d}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

Substitute the values: $T_1 = 273 + 300 = 573\text{K}$ $\quad$ $T_2 = 273 + 350 = 623\text{K}$

$D_1 = 7.8 \times 10^{-11}\ \text{m}^2/\text{s}$ $\quad$ $R = 8.31\ \text{J/(mol K)}$

$Q_D = 41{,}500\ \text{J/mol}$

$$D_2 = (7.8 \times 10^{-11}\ \text{m}^2/\text{s}) \exp\left[\frac{-41{,}500\ \text{J/mol}}{8.314\ \text{J/mol-K}}\left(\frac{1}{623\ \text{K}} - \frac{1}{573\ \text{K}}\right)\right] = 15.7 \times 10^{-11}\ \text{m}^2/\text{s}$$

Diffusion and temperature

Example 2: – Diffusion and activation energy

The diffusion coefficients for nickel in iron are given at two temperatures:

$T(K)$	$D(m^2/s)$
1473	2.2×10^{-15}
1673	4.8×10^{-14}

(a) Determine the values of D_0 and the activation energy Q_d.

(b) What is the magnitude of D at 1573 K?

Diffusion and temperature

Example 2 solution:

a) Using the equations for D as a function of 1/T, we set up two simultaneous equations with Q_d and D_0 as unknowns as follows:

$$\ln D_1 = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T_1}\right) \qquad \ln D_2 = \ln D_0 - \frac{Q_d}{R}\left(\frac{1}{T_2}\right)$$

Solving for Q_d in terms of temperatures T_1 and T_2 (1473 K and 1673 K) and D_1 and D_2 (2.2×10^{-15} and 4.8×10^{-14} m^2/s), we get

$$Q_d = -R\,\frac{\ln D_1 - \ln D_2}{\frac{1}{T_1} - \frac{1}{T_2}} = -(8.31\ \text{J/mol-K})\,\frac{\left[\ln(2.2 \times 10^{-15}) - \ln(4.8 \times 10^{-14})\right]}{\frac{1}{1473\ \text{K}} - \frac{1}{1673\ \text{K}}}$$

Q_d = 315,700 J/mol

Diffusion and temperature

Example 2 solution:

b) Solving for D_0 from a rearranged form of Equation for D_1 (and using the 1473 K value of D)

$$D_0 = D_1 \exp\left(\frac{Q_d}{RT_1}\right) = (2.2 \times 10^{-15}\ \mathrm{m^2/s}) \exp\left[\frac{315{,}700\ \mathrm{J/mol}}{(8.31\ \mathrm{J/mol\text{-}K})(1473\ \mathrm{K})}\right] = 3.5 \times 10^{-4}\ \mathrm{m^2/s}$$

Diffusion and temperature

<u>Example 3:</u> – Non-steady State Diffusion and temperature

An FCC iron-carbon alloy initially containing 0.20 wt% C is carburized at an elevated temperature and in an atmosphere that gives a surface carbon concentration constant at 1.0 wt%. If after 49.5 h the concentration of carbon is 0.35 wt% at a position 4.0 mm below the surface, determine the temperature at which the treatment was carried out.

Diffusion and temperature

Example 3 solution:

$$\frac{C(x,t)-C_o}{C_s-C_o}=1-\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- t = 49.5 h — x = 4 x 10^{-3} m
- C_x = 0.35 wt% — C_s = 1.0 wt%
- C_o = 0.20 wt%

$$\frac{C(x,t)-C_o}{C_s-C_o}=\frac{0.35-0.20}{1.0-0.20}=1-\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)=1-\operatorname{erf}(z)$$

$$\operatorname{erf}(z) = 0.8125$$

Diffusion and temperature

Example 3 solution:

Calculating the value of z for which the error function is 0.8125 by interpolation

z	erf(z)
0.90	0.7970
z	0.8125
0.95	0.8209

$$\frac{z-0.90}{0.95-0.90}=\frac{0.8125-0.7970}{0.8209-0.7970}$$

$$z = 0.93$$

Now solve for D

$$z=\frac{x}{2\sqrt{Dt}} \Longrightarrow D=\frac{x^2}{4z^2t}$$

$$D=\left(\frac{x^2}{4z^2t}\right)=\frac{(4 \times 10^{-3}\,\text{m})^2}{(4)(0.93)^2(49.5\ \text{h})}\frac{1\ \text{h}}{3600\ \text{s}}=2.6 \times 10^{-11}\ \text{m}^2/\text{s}$$

Diffusion and temperature

Example 3 solution:

Solve for the temperature at which D has the above value

$$T = \frac{Q_d}{R(\ln D_o - \ln D)}$$

The values for diffusion of C in FCC Fe

$D_o = 2.3 \times 10^{-5}\ \mathrm{m^2/s}$ $Q_d = 148{,}000\ \mathrm{J/mol}$ These were given in a previous problem

$$T = \frac{148{,}000\ \mathrm{J/mol}}{(8.314\ \mathrm{J/mol\text{-}K})(\ln 2.3\mathrm{x}10^{-5}\ \mathrm{m^2/s} - \ln 2.6\mathrm{x}10^{-11}\ \mathrm{m^2/s})}$$

$$\boxed{T = 1300\ \mathrm{K} = 1027^\circ\ \mathrm{C}}$$

Poll – Answers in WeChat

Q1) Assuming that the diffusion coefficient depend on temperature, how does the atomic size affect the diffusion coefficient?

A) Diffusion of smaller atoms increases the value of D;
B) Diffusion of larger atoms increases the value of D;
C) The size of the atom does not affect the value of D.

Poll – Answer

Q1) Assuming that the diffusion coefficient depend on temperature, how does the atomic size affect the diffusion coefficient?

A) Diffusion of smaller atoms increases the value of D.

Effect of diffusion species

Diffusing species	Host metal	D (m^2/s)
C	Fe (α-Fe)	2.4×10^{-12}
Fe	Fe (α-Fe)	3.0×10^{-21}
Cu	Cu	4.2×10^{-19}
Zn	Cu	4.0×10^{-18}

The magnitude of the diffusion coefficient is indicative of the rate at which atoms diffuse.

Interstitial diffusion is faster than self-diffusion mechanism.

Summary

Diffusion is **FASTER** for...

- Open crystal structures;
- Materials with secondary bonding;
- Smaller diffusing atoms;
- Lower density materials;
- Higher temperature.

Diffusion is **SLOWER** for...

- Close-packed structures of higher density materials;
- Materials with covalent bonding;
- Larger diffusing atoms.